

Using Model Mixtures enhanced with Textual Analyses to Forecast Return Distributions

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Motivation: Why Combine Models?

Proposal: mix distributions and estimate model weights and transition parameters. Why?

- Rather than assuming that one model will cover all possibilities
- Recognize the limited domain of particular models, e.g. good versus bad times may involve very different decisions and outcomes
- Identify the most appropriate model for the various ranges of potential outcomes relevant for one's decision
- Combine (mix) models with appropriate submodel or regime transition structure
 - Features of the component distributions improve estimates of the aggregate mixture
- Estimate submodel probabilities and model transition probabilities
- Update those probabilities and parameters as new information arrives (Bayesian learning)
- Powerful method of managing model uncertainty and associated model risk

Investigate whether decision-makers care or care differently about the different components

Outline

- This presentation will provide examples of theory-informed mixture models which can improve forecasts, providing better guidance for decisions.
- I will discuss the power of mixture models for capturing the dynamics (e.g., changing shape) of the distributions and summarize three applications
- The flexible non-linear time-series models that result can be disciplined with parametric structure derived from theory and sensible priors, distinguishing them from black-box nonlinear models.
- Fundamentals, which are treated as latent in the dynamic time-series models, can be used to better understand sources of the dynamics and forecast change points.
- For example, textual analyses converting qualitative information (e.g., news) to quantitative measures (e.g., sentiment) can be useful for capturing the dynamics of the mixture parameters.
- A theoretical representation of behaviour (a nonlinear pricing kernel derived from a generalized preference function that includes attitudes about both variance risk and skewness risk) can be used to evaluate to what extent participants care about the dynamics of the stock return distributions (tail risk is priced).

Example Applications of Mixtures

- *Application 1:* How useful are historical data for forecasting return distributions?
 - Bayesian averages of submodels which have different data supports
 - Identify and integrate out structural breaks for improved forecasts
 - Addresses the fundamental efficiency vrs bias problem
 - Identifies optimal data windows for correlation and return distribution forecasts

- *Application 2:* Can we forecast regime changes in stock market returns?
 - Mixture of bear market, bear rally, bull and bull correction states
 - A probabilistic approach to forecasting market dynamics
 - Use the entire distribution to help identify subregimes (bear rallies, bull corrections)

- *Application 3:* Are jumps in stock returns (tail risk) priced?

CPN distribution varies over time as directed by jump arrival & preferences

 - Mix a smooth (persistent) stochastic vol component with an outlier component
 - Include a jump-size distribution for jump reversals and heterogenous jumps
 - Time-varying jump arrival is the mixing parameter
 - News arrival can help explain dynamics of jump arrival and jump-size distribution.
 - Nonlinear pricing kernel captures preference components (including prudence)

- Forecasting with nonlinear mixture distribution parameterizations vs machine learning

App 1: Using Mixtures to Accommodate Structural Breaks

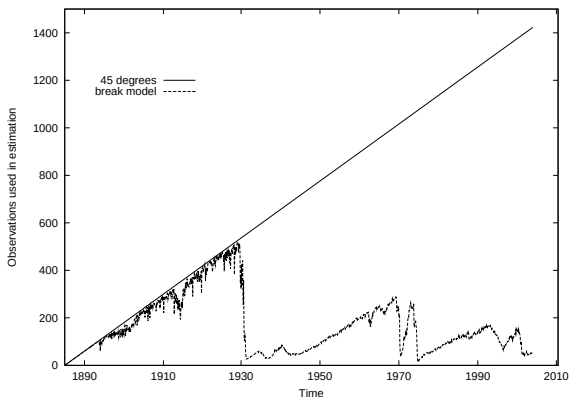
Focus on potential breaks as source of changes in the unconditional distribution of returns
Forecast distributions using a mixture of submodels – each with a different data support

- View the data as being partitioned (by the breaks) into regimes
- New submodels introduced periodically to check for breaks
- Estimate submodel probability weights and the probability of breaks in real time
- Bayesian average of submodels integrates out structural breaks for improved forecasts
- Optimal use of historical data (we show that fixed window is not optimal)
- Probabilistic approach to both past and future breaks
- We learn about breaks over time (time-varying submodel uncertainty)
- Submodel probability distribution provides estimate of most recent break dates
- Real-time distribution forecasts from our break model are superior to alternatives

Reference: 'How useful are historical data for forecasting the long-run equity return distribution?'
(with Maheu), *Journal of Business & Economic Statistics*, 27 (1), pp. 95-112.

Optimal Data Windows

- The expected number of useful observations for forecasting at each t



- Note that the optimal data window can change at each t
- Also applied this mixture approach to correlation forecasts

Application 2: Forecasting Stock Return Distributions

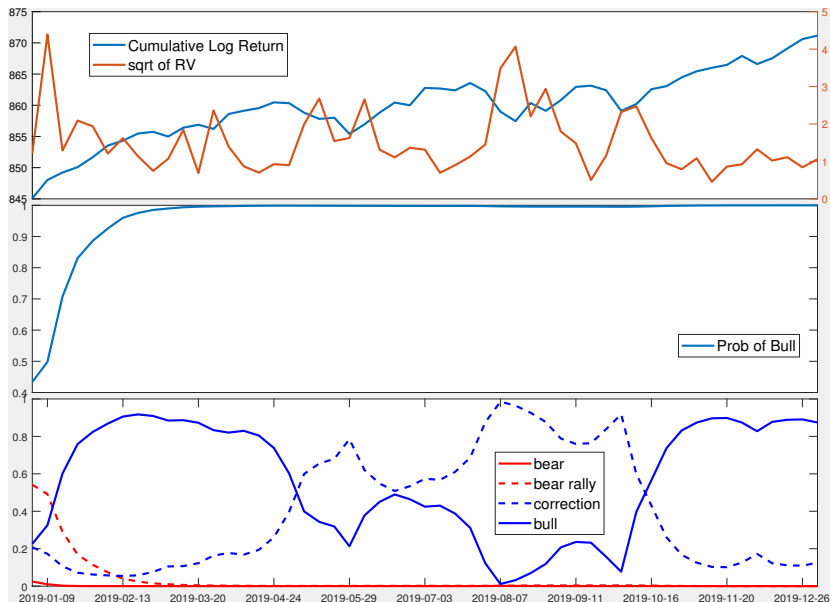
Nonlinear model (4 latent states with sparse transition probability structure): generates probability estimates for market states and distribution forecasts for aggregate stock returns

- Include bear rallies and bull corrections to capture heterogeneous intra-regime dynamics
 - Allows transitions from a bear rally either to a bull or back to the bear state
 - Allows transitions from a bull correction to a bear or back to the bull state
- Provides a full probability model of stock market phases
 - The state-specific distributions provide useful information
 - The mixture governs the market dynamics
- Can forecast (market states, risks, and returns) out-of-sample
 - Computation of the expected return takes all 4 possible future states into account
 - State dependent means and other parameters are components of those calculations
 - Conditional VaR predictions are sensitive to market regimes
- Bayesian estimation accounts for parameter and regime uncertainty

References:

- 'Identifying Bull and Bear Markets in Stock Returns', (with Maheu), *Journal of Business & Economic Statistics* 18 (1), pp.100-112.
- 'Components of Bull and Bear Markets: Bull Corrections and Bear Rallies' (with Maheu and Song), *Journal of Business & Economic Statistics* 30 (3), pp.391-403.
- 'Bull and bear Markets during the COVID-19 pandemic', (with Maheu and Song), *Finance Research Letters* Vol 42, 21 pages.

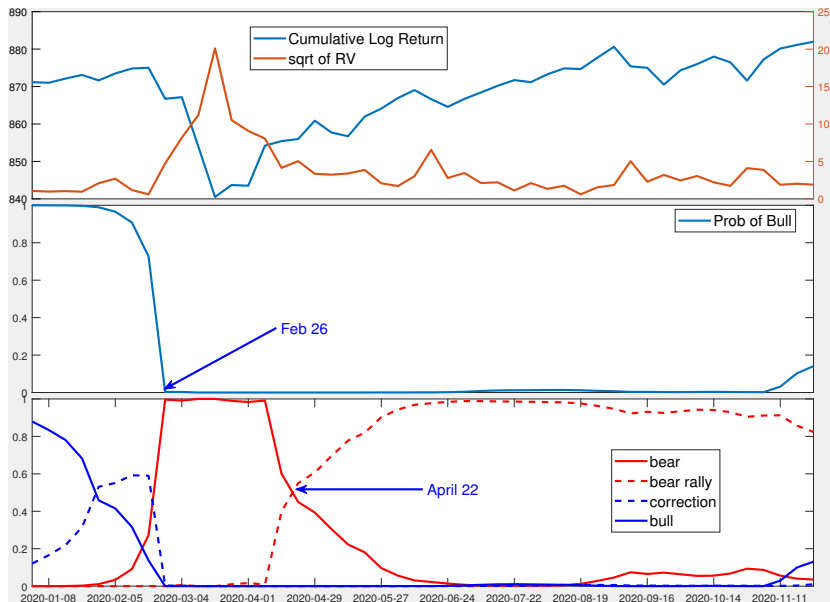
Smoothed Estimates 2019



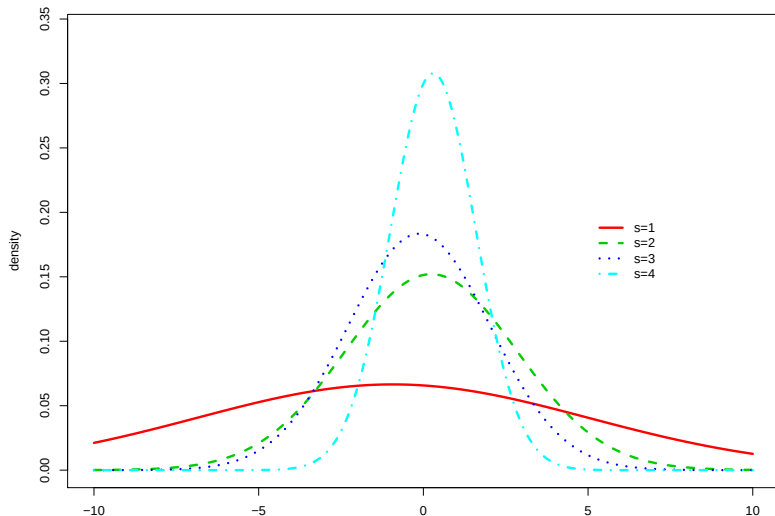
Weekly Stats, Feb-May 2020

week end	wk ret	wk vol	YTD ret	latent state probabilities			
				bear	bear rally	bull correction	bull
1/2/2020	0.55%	1.06%	0.55%	0.000	0.000	0.171	0.829
1/8/2020	-0.15%	0.97%	0.40%	0.000	0.000	0.218	0.781
1/15/2020	1.11%	1.03%	1.51%	0.001	0.000	0.244	0.755
1/22/2020	0.98%	0.96%	2.49%	0.003	0.001	0.352	0.645
1/29/2020	-1.47%	2.09%	1.03%	0.011	0.001	0.616	0.373
2/5/2020	1.86%	2.70%	2.88%	0.040	0.001	0.562	0.398
2/12/2020	1.33%	1.17%	4.22%	0.097	0.002	0.581	0.320
2/19/2020	0.20%	0.61%	4.41%	0.265	0.002	0.618	0.115
2/26/2020	-8.30%	4.74%	-3.89%	0.998	0.000	0.002	0.000
3/4/2020	0.44%	8.16%	-3.45%	0.994	0.005	0.001	0.000
3/11/2020	-13.26%	11.21%	-16.71%	1.000	0.000	0.000	0.000
3/18/2020	-13.38%	20.10%	-30.09%	1.000	0.000	0.000	0.000
3/25/2020	3.18%	10.51%	-26.91%	0.987	0.013	0.000	0.000
4/1/2020	-0.20%	9.07%	-27.11%	0.982	0.018	0.000	0.000
4/8/2020	10.72%	8.05%	-16.40%	0.996	0.005	0.000	0.000
4/15/2020	1.21%	4.14%	-15.19%	0.657	0.342	0.000	0.001
4/22/2020	0.57%	5.04%	-14.62%	0.528	0.470	0.000	0.001
4/29/2020	4.89%	3.35%	-9.73%	0.487	0.512	0.002	0.000
5/6/2020	-3.15%	3.23%	-12.88%	0.413	0.586	0.002	0.000
5/13/2020	-1.00%	3.39%	-13.88%	0.338	0.649	0.002	0.011

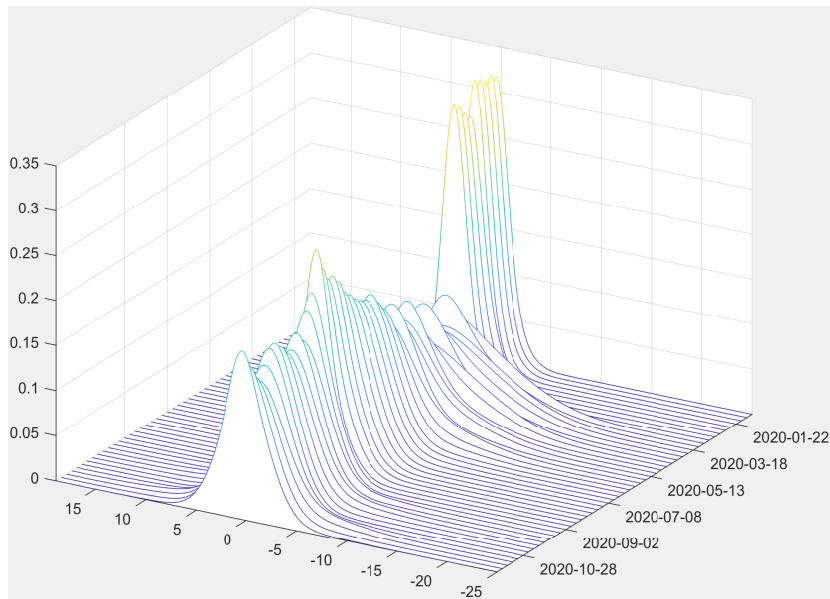
Smoothed Estimates 2020



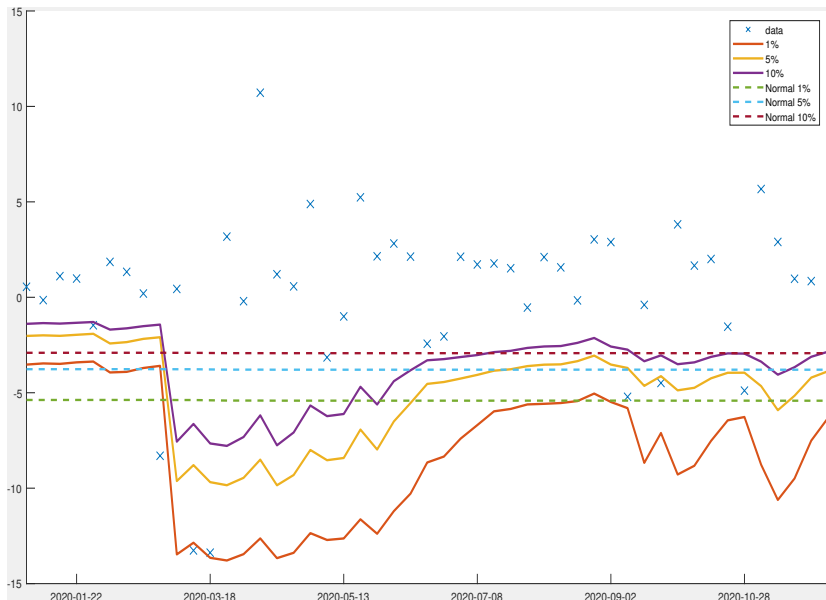
Distributions for Alternative Market States



Out-of-sample: Predictive Density 1 week ahead



Out-of-sample: Value-at-risk 1 week ahead



Investment Returns in 2020

	Return	Sharpe Ratio	number of transactions
Strategy B ^a : $\tau_B = 0.5$	-0.009	-0.048	3
Strategy S ^b : $\tau_S = 0.5$	0.220	1.203	4
Buy-and-hold	0.131	0.566	0

The returns are annualized. Transaction costs are 0.3 cents per share.

^a Buy if $P(B_t = 2 \mid r_{1:t-1}) > \tau_B$ and sell otherwise.

^b Buy if $P(s_t = 2 \mid r_{1:t-1}) > \tau_S$ or $P(s_t = 4 \mid r_{1:t-1}) > \tau_S$. Sell otherwise.

Summary: Model Mixtures for Forecasting Stock Returns

- Parameterized a 4-state Markov-switching model for stock returns
 - Offers richer characterizations of market dynamics
 - Two states govern the bear regime
 - Two states govern the bull regime
 - Intuitive restrictions on the state transition probabilities improves forecasts
- Heterogeneous regimes and intra-regime dynamics
 - Allow for bear rallies and bull corrections without a regime change
 - Most regime turning points occur through bear rally or bull correction
 - Volatility dynamics within regimes is important for transitions
 - investors care about both risk and return
- Asymmetric transitions both within regimes and between regimes
 - Bull corrections revert to bull more often than bear rallies bounce back to bear
 - Realized bull and bear regimes can be very different over time
- Probability statements on regimes and future returns available
- Bull corrections and bear rallies empirically important
- Realized bull and bear regimes can be very different over time
- VaR predictions sensitive to market regimes

Motivation Revisited

Can regime probabilities improve risk management & investment decisions?

- Using probabilities to anticipate state or regime transistions could be important for:
 - Investment decisions: rebalance portfolios – e.g. 'take risk off'
 - Risk management decisions: VaR, hedging, etc.

What causes dynamics of the regimes?

- Stay invested and 'weather the storm' for normal fluctuations in returns and risk
- If a bull correction persists and gets deep enough
 - Investors may voluntarily re-allocate portfolios to 'take risk off'
 - Or be forced to do so due to margin calls or factors that kick-in at a threshold
 - These responses could re-inforce the correction & push the market into the bear state with its own reinforcing persistence

Could a nonlinear pricing kernel generate these dynamics?

- A Preference Regime Model of Bull and Bear Markets (Gordon & St-Amour, AER)
- Skewness risk premia based on generalized preferences (Maheu & McCurdy & Zhao, JFE)

Application 3: Pricing Jump (Tail) Risk

Jumps are large price changes:

- Jumps matter for risk management, portfolio selection, pricing, etc.
- We filter returns into large versus smaller changes simultaneously with estimation of all the parameters of the conditional distribution

Are jumps a source of changes in the shape of return distributions?

- Jumps contribute to the conditional variance and drive dynamics in the tail

Do investors care about both the dispersion of their wealth & tail risk for wealth?

- That is, do jumps contribute to the equity premium?

Reference: 'Do jumps contribute to the dynamics of the equity premium?

(with Maheu and Zhao), *Journal of Financial Economics*, Vol 110 (2), pp. 457-477..

Dynamics of Higher-Order Moments

In this paper changes in higher-order moments include:

- Dynamics of the conditional variance driven by
 - The direct effect of jumps (jump arrival and jump size)
 - A 2-component asymmetric GARCH process which allows persistence of jumps to be different than normal innovations

Special cases:

- If there are no jumps, conditional skewness is 0 and conditional kurtosis is 3
- If jump arrival is positive but constant, 3rd and 4th moments are non-normal but constant

In our parameterization with time-varying jump arrival:

- All 4 moments of the distribution are time varying

Model Component 1/5: Diffusive Innovation

► excess return dynamics

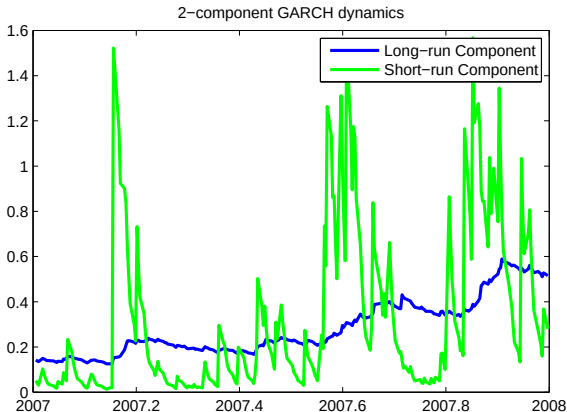
Total return innovation: $\epsilon_{t+1} \equiv \epsilon_{1,t+1} + \epsilon_{2,t+1}$

Diffusive innovation: directed by 2-component GARCH dynamics

$$\epsilon_{1,t+1} | \Phi_t \sim N(0, \sigma_t^2)$$

$$\sigma_t^2 = \text{long-run component} + \text{short-run component}$$

⇒ Captures both **smooth/long-run** component and **transitory/short-run** component



Model 2/5: The volatility clustering directing process

- The normal innovation to returns, $\epsilon_{1,t+1}$, is assumed to be directed by a 2-component GARCH specification with feedback from jumps

$$\begin{aligned}\epsilon_{1,t+1} &= \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim NID(0, 1), \quad \epsilon_{1,t+1} | \Phi_t \sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \sigma_{1,t}^2 + \sigma_{2,t}^2 \\ \sigma_{1,t}^2 &= \omega + g_1(A_1, \Phi_t) \epsilon_t^2 + \beta_1 \sigma_{1,t-1}^2 \\ \sigma_{2,t}^2 &= g_2(A_2, \Phi_t) \epsilon_t^2 + \beta_2 \sigma_{2,t-1}^2 \\ \epsilon_t &= \epsilon_{1,t} + \epsilon_{2,t}\end{aligned}$$

- The generalized **news impact coefficient** for the i^{th} GARCH component is

$$\begin{aligned}g_i(A_i, \Phi_t) &= \exp(\alpha_i + I(\epsilon_t)(\alpha_{a,j,i} E[n_t | \Phi_t] + \alpha_{a,i})), i = 1, 2 \\ I(\epsilon_t) &= 1 \text{ if } \epsilon_t < 0, \text{ otherwise } I(\epsilon_t) = 0\end{aligned}$$

- Note that this parameterization incorporates:
 - Asymmetric impact from good versus bad news
 - Asymmetric feedback to future vol from jumps vs normal innovations

Model Component 3/5: Jump Innovation

Total return innovation: $\epsilon_{t+1} \equiv \epsilon_{1,t+1} + \epsilon_{2,t+1}$

Compensated jump innovation: directed by compound poisson distribution

$n_{t+1} \equiv$ Number of jumps, $Y_{t+1,k} \equiv$ Size of one jump
 $n_{t+1} | \Phi_t \sim \text{Poisson}(\lambda_t), \quad Y_{t+1,k} \sim N(\theta, \delta^2)$

$$\begin{aligned}
 P(n_{t+1} = j | \Phi_t) &= \frac{\exp(-\lambda_t) \lambda_t^j}{j!}, \quad j = 0, 1, 2, \dots \\
 \Rightarrow \lambda_t &\equiv E[n_{t+1} | \Phi_t]
 \end{aligned}$$

$$\begin{aligned}
 \epsilon_{2,t+1} &= J_{t+1} - E[J_{t+1} | \Phi_t] \\
 &= \sum_{k=1}^{n_{t+1}} Y_{t+1,k} - \theta \lambda_t, \\
 \Rightarrow E[\epsilon_{2,t+1} | \Phi_t] &= 0
 \end{aligned}$$

Model Component 4/5: Jump Arrival Dynamics

Jump arrival parameterization:

$$\lambda_t = \gamma_0 + \gamma_1 \lambda_{t-1} + \gamma_2 \zeta_t$$

- The jump arrival *innovation* is defined as

$$\begin{aligned} \zeta_t &\equiv E[n_t | \Phi_t] - \lambda_{t-1} \\ &= \sum_{j=0}^{\infty} j P(n_t = j | \Phi_t, \Theta) - \lambda_{t-1} \\ &= E[n_t | \Phi_t] - E[n_t | \Phi_{t-1}] \end{aligned}$$

- That is, the update in the expected number of jumps for period t when information is updated from $t-1$ to t
- Jumps are latent: $P(n_t = j | \Phi_t, \Theta)$ is obtained from the estimation filter

Jump arrival has its own dynamics, not necessarily the same as the volatility dynamics

Many models in the literature have specified:

- Time-varying jump arrival (jump intensity) as a function of the diffusive volatility
For example, $\lambda_t = \gamma_0 \sigma_t^2$
- We will provide some comparisons below

Model Component 5/5: Specification for Equity Premium

Our maintained parameterization for the market equity premium is:

- $m_t \equiv$ conditional market equity premium: estimated with & without an intercept μ

$$m_t = \mu + \psi_v v_t + \psi_s s_t + \psi_k k_t$$

- $v_t \equiv$ conditional variance; $s_t \equiv$ conditional skewness; $k_t \equiv$ conditional kurtosis
- Decreasing absolute prudence (Kimball, Econometrica) implies that: $\psi_s \leq 0, \psi_k \geq 0$

In addition to alternative versions discussed in the robustness section, we also estimate

- 1 An unrestricted parameterization of this conditional equity premium; and
- 2 The special case of power utility (Duan and Zhang)

$$m_t = \mu + \psi_v v_t + \psi_s v_t^{\frac{3}{2}} s_t + \psi_k v_t^2 k_t,$$

in which $\psi_v = (\gamma - 0.5)$, $\psi_s = -(3\gamma^2 - 3\gamma + 1)/6$, $\psi_k = (4\gamma^3 - 6\gamma^2 + 4\gamma - 1)/24$ where γ is defined as the CRRA

► derivation

Results 1/5: Higher-order Moments in the Equity Premium

► Likelihood Function

$$m_t = \mu + \psi_v v_t + \psi_s s_t + \psi_k k_t$$

Parameter	Pricing Models			
	prudence	variance	skewness	kurtosis
μ		0.020 [3.337]	-0.041 [-2.248]	-0.022 [-1.194]
ψ_v	0.026 [3.396]	0.022 [2.216]	0.045 [4.052]	0.034 [3.151]
ψ_s	-0.027 [-3.423]		-0.069 [-3.015]	
ψ_k	0.000 [0.000]			0.005 [2.295]

- Both variance and skewness are significantly priced
- At the market level skewness and kurtosis pick up the same source of risk
 - Also evident from high negative correlation (-96%) between skewness and kurtosis

Results 2/5: Importance of Three Parts in the Model

1. Time varying jump arrival is important
2. 2-component diffusive volatility dynamics is important

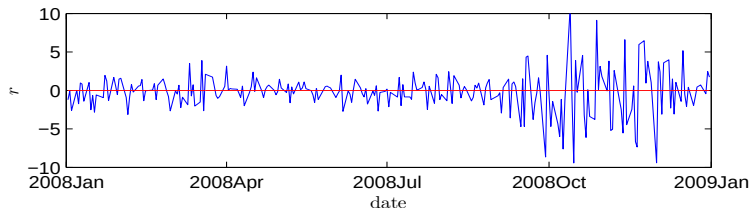
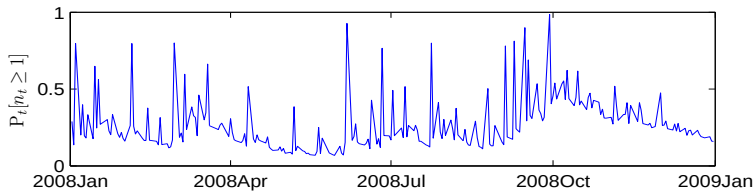
Null hypothesis	LRT Statistic	P-value	Higher moments priced?
Constant jump intensity	83.8	6.34×10^{-19}	No
1-component diffusive volatility	289	2.60×10^{-61}	No

3. Nonlinear pricing of jump intensity is important:

Parameter	Pricing Models	
	our model	linear: $m_t = \psi_v \sigma_t^2 + \psi_j \lambda_t$
ψ_v	0.026 [3.40]	0.017 [1.44]
ψ_s	-0.027 [-3.42]	
ψ_k	0.000 [0.00]	
ψ_j		0.188 [2.47]
lgl	-26386.79	-26392.30

Results 3/5: Diffusive vs Jump Components of the Variance

Sample period	$\bar{\lambda}$	$\bar{v}$	v due to jump part	v due to diffusive part
1926:01–2011:12	0.136	1.054	27%	73%
1926:01–1940:12	0.161	2.130	19%	81%
2008:01–2011:12	0.199	2.813	15%	85%
2008:09–2009:01	0.308	10.167	4%	96%



Results 4/5: Contribution of Jumps to the Equity Premium

Our **time-series** approach with daily **index** returns generates:

- Contribution of jumps to the equity premium about **3.61%** on average
 - Approximately one-third (1.06% per annum) of the jump risk premium comes through contribution of jumps to variance dynamics

$$\psi_v \times (\bar{\lambda} \times (\theta^2 + \delta^2))$$
 - The remainder (2.55% per annum) of the jump risk premium comes through contribution of jumps to skewness
- Skewness premium ($\psi_s \times E[s_t]$) of **3.4%** on average

Results 5/5: Out-of-Sample Asset Allocation

Evaluate the out-of-sample portfolio performance with different equity premium models

Utility calibrated to our model and parameter estimates

- Consistent with our assumed pricing kernel

Table: Out-of-Sample Portfolio Performance

Model	Realized Utility		CEQ
	Mean	Std	
Prudence	3.34E-03	5.05E-02	4.005
Variance	3.09E-03	5.73E-02	3.719
GIM-1	2.88E-03	5.51E-02	3.469
Constant	2.80E-03	5.70E-02	3.375

- Realized utility and CEQ improve when pricing higher-order moments (jumps)

Summary: Component Mixtures for Pricing Jump Risk

We show that jumps are priced through higher-order moments in the market return

- More general preferences
- Time-varying jump arrival with own price dynamic
- 2-component diffusive volatility, and
- nonlinear pricing of jump intensity

are **all** important for identifying market prices of **diffusive and jump** risks

Empirically, for our sample, we find:

- Jump risk premia of 3.61% per annum
- Skewness and kurtosis pick up the same tail risk at the market level
Skewness premium of 3.4% per annum
- Pricing higher-order moments is important for understanding the relationship between equity premium and conditional variance
- Out-of-sample portfolio performance improves when pricing jumps/higher-order moments

Using Textual Analyses of News to Augment Forecasts

Extension: replace latent dynamics by textual analyses of high frequency news arrival

Reference: 'News as sources of jumps in stock returns: Evidence from 21 million news articles for 9000 companies' (with Jeon and Zhao), *Journal of Financial Economics*, Vol 145, pp. 1-17.

- We use a large sample: 21+ million news articles for 9,000+ firms
- Using textual analysis, we quantify frequency and content (tone and uncertainty) of daily firm-specific news and test relationship with jumps in those individual stock returns

Results: Occurrence of jumps & the jump-size distribution for stock returns are significantly related to news frequency & content of firm-specific news

- Jump probability is positively and significantly related to the frequency of news arrival
- The variation in jump-size distributions is related to news tone and uncertainty

Does incorporating quantifiable measures of firm-specific news flow improve nonlinear time-series forecasts of the distribution of individual stock returns?

- The news-enhanced model outperforms (with statistically significant coefficient loadings) the benchmark model for most firms
- Applications for using observable news flow in option valuation and risk management

Discussion: forecasting with nonlinear mixture distribution parameterizations vs machine learning